

The Plausibility of Exobiology on HD-127688. It is highly plausible given HD-127688's advantageous physical and orbital properties. According to our calculated values for the data, HD-127688 possesses a mass of 1.4 Earth masses and a radius of 1.12 Earth masses, with an extremely dense 5.5 g/cm; this density is virtually identical to the density of Earth at 5.52 g/cm and we can therefore consider HD-127688 to be a terrestrial, rocky planet and specifically as a super-earth, since its mass and radius are larger than Earth's. In addition, the planet orbits a K-type star with a radius of 1.12 Earth radii with an orbital distance of 0.8 AU. The habitable zone for a K-type star (according to established planetary science) ranges between 0.5 AU and 1.1 AU and since 0.8 AU lies comfortably between 0.5 and 1.1 AU, the planet rests within the habitable zone. Such an orbital distance coupled with a rocky surface likely translates to the existence of liquid surface water which, as the most crucial solvent required for life, along with the exceedingly stable life of K-type stars (18 to 49 Gyr) makes HD-127688 a highly favorable target to study the existence of life.

State of Exobiology at 3.5 billion Years Old (The Proterozoic Eon) As the exact age of the system is unknown, I will examine life on the planet as if it is 3.5 billion years old (3.5 Gyr). If we assume that the timescales and chemistry of the development of life are roughly consistent with the development of life on our own planet, then life on HD-127688 at 3.5 Gyr is mid-Proterozoic eon. Therefore, biology on HD-127688 will have moved well beyond the initial emergence of life, but will not yet have become intelligent. Single celled life forms first arose long before in the Archaean eon, so at 3.5 billion years in the Proterozoic, biology on HD-127688 will be seeing the success of eukaryotic cells which, having become far more advanced and nucleated, will be abundant in its oceans. Due to the rise of multicellular life in the Proterozoic eon, it is likely that rudimentary, simple multicellular organisms will be flourishing within the oceans of HD-127688, such as early sponges or soft macroscopic organisms (though not intelligent life – biological and advanced intelligence arise far later in a planet's history during the Phanerozoic and Eozoic eons respectively). It is therefore likely, given its age, that life on HD-127688 at 3.5 billion years would be well advanced in its biological complexity, and a rich tapestry of eukaryotic cells and the earliest multicellular life forms would exist there; as an age and time at which life has

become more advanced than mere simple life forms, HD-127688 at 3.5 Gyr offers a highly suitable opportunity to study the potential of exobiology.